	Sep 29–Oct 2 CPU transmitter	Oct 14 limits, LUT DMA	Oct 15 a.m. 16 Msym/s, 8PSK	Oct 15 midday APSK	Oct 15 p.m. R&D round 1	Oct 15 eve. round 2	Oct 15–16 round 3
Pulse shaping	○ LUT kernels, 918 → 10.8 cycles		○ 2 SPS (images −18 dBc) ○ signed-plane 8PSK / VCM	● APSK on the CPU	● four-level 8PSK table ✕ APSK table in hardware (MiB)		● 8-bit table rows
DMA/PIO output		○ LUT rows by DMA	○ PIO addresses, cyclic		● PIO addresses, live, 8 Msym/s ✕ same at 16 Msym/s: stalls	● DMA read priority: 16 Msym/s ✕ four-writer router (A2)	✕ 1 chain per axis at 16 Msym/s ○ 4 chains, SRAM addresses ● 4 PIO producers, live
FEC and mapping	● encoder v1 → v4, 4.1M → 237k				● streaming LDPC packer ✕ scheduled LDPC: slower live	● t=8 BCH, QPSK 9/10	○ two-core frame-parallel FEC ● C4 kernels, −11 to −14% ✕ 8PSK FEC at 16 Msym/s
Memory, clock, bus			○ BCH in SCRATCH_Y, −31%		● GPIO28 DDR clock	● core-0 stack overflow fixed	◇ DMA row loop 10–18 cycles ◇ two cores clash in SCRATCH_Y ○ self-re-arming DMA grants ✕ 256 MHz overclock
Higher order			○ 8PSK and VCM at 150 MHz	● 16APSK / 32APSK complete	◇ APSK below QPSK-16		● 8PSK complete, 8 Msym/s ● 8PSK waveform, 16 Msym/s
Input	● PV-SPI from a CM5, 210 s		● PIO/DMA SPI ingress, 32 MHz				✕ USB input, hardware FEC

● current best ○ worked, superseded or not adopted ◇ measurement that explained a limit ✕ failed or ruled out